

Jinwei Liu

Ph.D. Student in Control Science and Engineering

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👤 Jinwei-Liu

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Profile

- Ph.D. student in Control Science and Engineering at the University of Science and Technology of China through a master's-to-Ph.D. transfer, advised by Prof. Yun-Bo Zhao in the Networked Intelligent Control Lab (NICLab).
- During my master's study, I worked with Pengfei Li, Associate Researcher at USTC.
- Research interests include human-machine hybrid intelligence, human-robot interaction, shared autonomy, and embodied intelligence.
- Current work focuses on shared control, teleoperation, reinforcement learning, and human-in-the-loop reinforcement learning.
- Interested in both human-to-machine intent communication and machine-to-human information expression.

Experience

Dobot, Embodied Intelligence Algorithm Intern

Shanghai, China
June 2026 – present
2 months

Education

Doc- University of Science and Technology of China, Control Science and Engineering
tor of • Master's-to-Ph.D. transfer; combined master's-doctoral track
Phi-
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Hefei, China
June 2026 – present

Mas- University of Science and Technology of China, Artificial Intelligence
ter of • Weighted average: 91.09; GPA: 3.99/4.3
Engi-
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Hefei, China
Sept 2024 – June 2026

Bach- Ocean University of China, Automation
elor of • Rank: 1/89; Weighted average: 92.42; GPA: 3.82/4.0
Engi- • Honors: National Scholarship (twice); Honorary Bachelor's Degree of Ocean Uni-
neer- versity of China
ing

Qingdao, China
Sept 2020 – June 2024

Publications

Delay-Aware Shared Control for Teleoperation Systems: Intent Trajectory Prediction Under Communication Latency

2026

Journal article on delay-aware shared control for teleoperation systems, using DA-RT to predict operator intent under communication latency.

Jinwei Liu, Pengfei Li, Yaqing Zhou, Tao Wang, Yu Kang, Yun-Bo Zhao

ieeexplore.ieee.org/document/11535330

Legible Shared Autonomy: Implicit Communication of Robot Belief through Motion Accepted paper on legible shared autonomy, enabling robot assistive motion to communicate inferred user goals and confidence through motion. Jinwei Liu, Pengfei Li, Shaofeng Chen, Tao Wang, Yun-Bo Zhao	2026
LSTM-MPC: Physically Feasible and Interpretable Trajectory Prediction Accepted paper on physically feasible and interpretable trajectory prediction via latent MPC parameter estimation. Jinwei Liu, Pengfei Li, Qianqian Zhang, Yunbo Zhao, Yu Kang	2026
Human-AI Shared Control via Motor Knowledge Neuron Stimulation Accepted paper on human-AI shared control through distributed motor knowledge neuron assemblies. Jinwei Liu, Pengfei Li, Sibin Huang, Yun-Bo Zhao, Yu Kang	2026